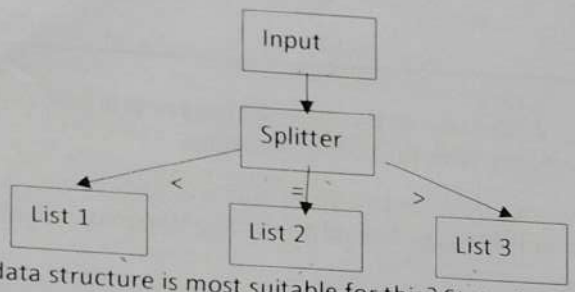
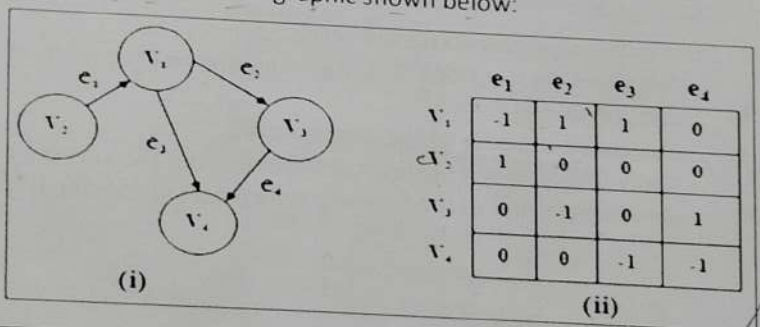
NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END-SEMESTER EXAMINATION – ODD SEMESTER 2025

III Semester CS34104 – Data Structures Max. Marks: 60

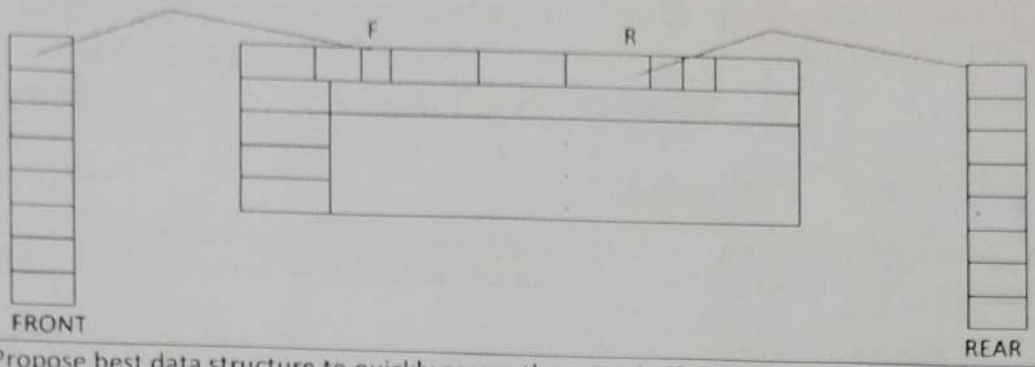
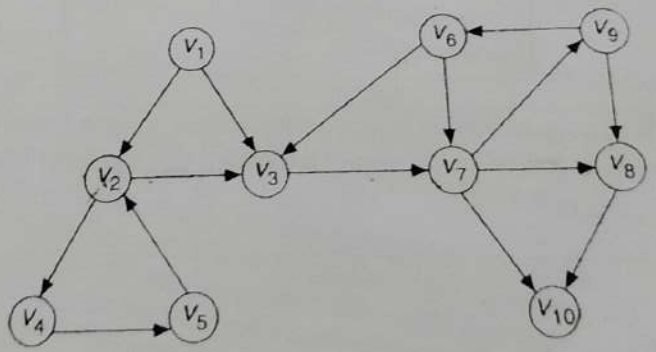
Answer all questions

Q. No	Question	Marks	CO & BL
1.	a) Write an algorithm to move zeros at one side of your preferred data structure. Solution should be in place. Analyze Best case, worst case and average case of the algorithm.	6M	CO-2 & Design
	b) You have given two doubly linked lists, $L1$ and $L2$; In both lists every node stores one integer value. Write an Algorithm named $SetDifference\ (List\ L1, List\ L2)$ that deletes all the nodes from $L1$ whose node element (i.e., integer value) matches with an element of some node in $L2$ (i.e., after running your algorithm, list $L1$ should not contain any node whose value is present in some node in $L2$). Mention the time complexity of your algorithm by assuming $L1$ contains m nodes and $L2$ contains n nodes.	6M	CO-1 & Design
2.	a) Write an algorithm to implement figure given below:  Which data structure is most suitable for this? State the reason with proof.	5M	CO-3 & Design
	b) Write an algorithm to create Adjacency Matrix from Incidence Matrix. For example, Incidence Matrix of directed graphic shown below:  (i)	5M	

	e_1	e_2	e_3	e_4
V_1	-1	1	1	0
V_2	1	0	0	0
V_3	0	-1	0	1
V_4	0	0	-1	-1

(ii)

3.	a) Explain how to modify Dijkstra's algorithm so that if there is more than one shortest path from u to v then it will choose that path which has least number of edges.	5M	CO-3 & Apply

	b) A priority queue can be implemented using a matrix P and two arrays $FRONT$ and $REAR$ pointing the front and rear of the queue. Obtain the various operations of a priority queue with such implementation.  	7M	CO-2 & Apply
4.	Propose best data structure to quickly access the records of cricket players in the world. These records are stored in file structure as shown in figure Name Country Total Run Number of 50's Number of 100's wickets collected Ranking R_1 R_2 R_3 . . R_n Suppose indexing is based on Ranking. Write steps of the algorithm to perform insertion, deletion of records. Show effect of each step with the help of example.	8M	CO-4 & Evaluate
5.	a) The radius of a tree is maximum distance from root to a leaf. Given a connected undirected graph, write an algorithm for finding for finding a spanning tree of minimum radius.	5M	CO-3 & Design
	b) Write the steps to efficiently store and retrieve the elements of a sparse matrix by using array. Suggest more suitable data structure for sparse matrix implementation with proper justification.	5M	CO-2 & Design
6.	With reference to the following graph do the following: I. Apply DFS on the graph to obtain finishing time of each vertex (start with vertex v_1) II. Compute transpose of the given graph III. Apply DFS on result of step II, where consider the vertices in the decreasing order of the finishing time (obtained in step I)  	8M	CO-3 & Apply

********** ALL THE BEST **********